

01. Introducció i Instal·lació de l'entorn de desenvolupament React Native

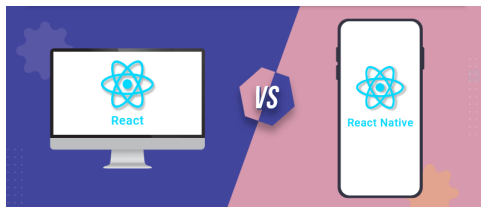
Xavi Lara - Profesor I.Sabadell xlara@ies-sabadell.cat



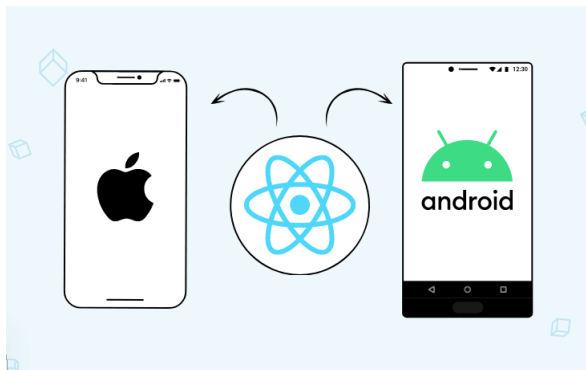
Institut Sabadell

Que es React Native?

React Native es un marc de desenvolupament (*framework*) de codi obert creat per Facebook, que permet construir **aplicacions mòbils para iOS i Android** utilitzant llenguatge **JavaScript** de forma molt més ràpida i eficient.



Es basa en la biblioteca de React per JavaScript que originalment va ser dissenyada per construir UI interactives en aplicacions web.



Avantatges pricipal de React Native

Permet crear aplicacions mòbils utilitzant un únic conjunt de codis base en lloc de tenir que escriure codis separats per iOS i Android, utilitzant components de UI que es tradueixen a elements natius de cada plataforma.

Característiques de React Native

- Permet compartir codi entre plataformes.
- Té un únic llenguatge de programació, simplificant el procés de desenvolupament.
- Ofereix una UI consistent en diferents dispositius, brindant experiència uniforme.
- Destaca per l'estabilitat i seguretat.
- S'integra de forma eficient amb molt complements y funcions de tercers.
- Es pot instal·lar amb NPM (*node package manager*), simplificant el procés de inici.
- Facilita la visualització immediata de canvis durant el desenvolupament.

Bridge de React Native

S'encarreguen de enllaçar el codi de la màquina virtual de JavaScript amb las API i el propi codi de la plataforma on s'executa l'app

La Máquina Virtual de React Native

JavaScriptCoreVM es la màquina virtual de JavaScript que utilitza React Native per a poder executar el codi en diferents entorns. Encara que aquesta execució en una màquina virtual fa que les **apps de React Native no siguin 100% natives**, sí es cert que consegueix una execució amb un rendiment similiar gracies a l'ús de components natius, optimització de processos i renderitzats, i execució de elements en según pla, entre altres funcionalitats.

0. Introducció a React Native

1. Instal·lació de React Native
2. Dispositius On Desenvolupar Aplicacions

0.1. Què es React Native?

0.2. Showcase de React Native



React Native is shaping mobile, web, and desktop experiences within Meta's product ecosystem, from Facebook Marketplace, Messenger Desktop, Ads Manager to the Meta Quest app and many more.



Facebook

[iOS · Android](#)



Facebook Ads
Manager

[iOS · Android](#)



Meta Quest

[iOS · Android](#)



Messenger Desktop

Desktop

[Learn more](#)



Microsoft leverages the power of React Native to deliver excellent customer experiences in some of its most well known apps.

Microsoft doesn't stop at mobile platforms either – Microsoft leverages React Native to target desktop too! Find out more in the [dedicated showcase](#) for React Native Windows and macOS.



Microsoft Office

[iOS · Android](#)

[Learn more](#)



Microsoft Outlook

[iOS · Android](#)

[Learn more](#)



Microsoft Teams

[iOS · Android](#)

[Learn more](#)



Xbox Game Pass

[iOS · Android](#)

[Learn more](#)



Skype

[iOS · Android](#)

[Learn more](#)



Amazon has used React Native to rapidly deliver new customer-facing features in some of its most popular mobile applications as early as 2016. Amazon also uses React Native to support customer-favorite devices such as the Kindle E-readers.



Amazon Shopping

[iOS · Android](#)



Amazon Alexa

[iOS · Android](#)



Amazon Photos

[iOS · Android](#)



Amazon Kindle

[Learn more](#)



Amazon Appstore

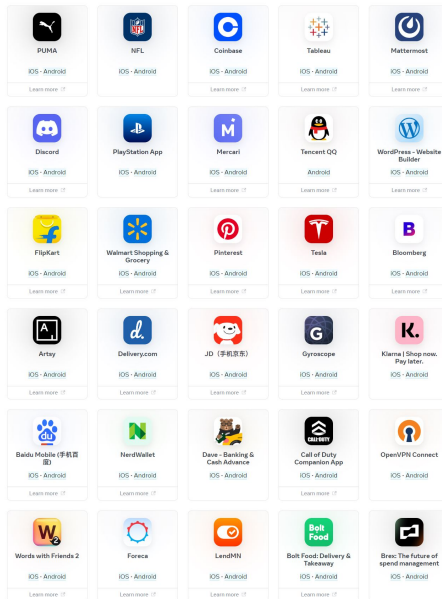
[Learn more](#)

0. Introducció a React Native

1. Instal·lació de React Native
2. Dispositius On Desenvolupar Aplicacions

0.1. Què és React Native?

0.2. Showcase de React Native



Instal·lació de React Native

[Learn](#) [About](#) [Download](#) [Blog](#) [Docs](#) [Certification](#) [Ctrl K](#) [🔍](#) [🔗](#) [🔄](#)

Run JavaScript Everywhere

Node.js® is a free, open-source, cross-platform JavaScript runtime environment that lets developers create servers, web apps, command line tools and scripts.

[Download Node.js \(LTS\)](#)

Downloads Node.js v20.18.0 with long-term support. Node.js can also be installed via package managers.

Want new features sooner? Get [Node.js v22.5.0](#) instead.

```
Create an HTTP Server | Write Tests | Read and Hash a File | Streams Pipeline | Work with Threads

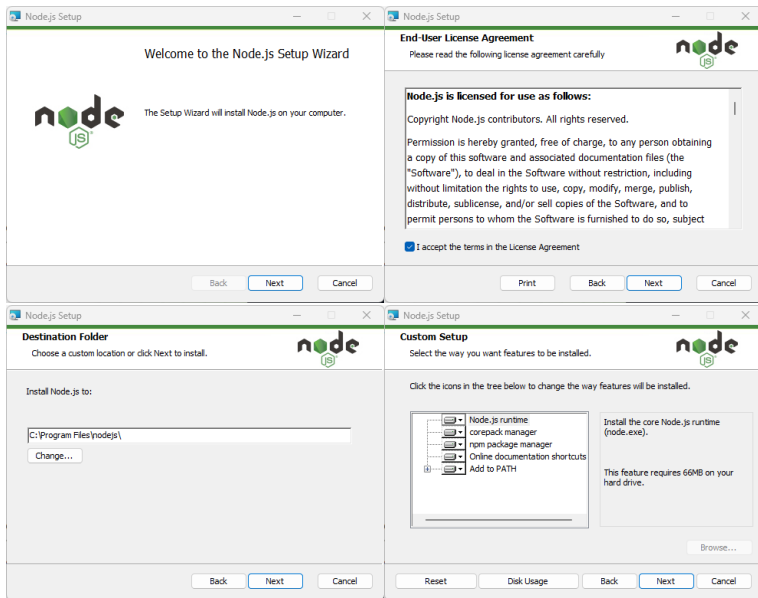
1 // server.js
2 import { createServer } from 'node:http';
3
4 const server = createServer((req, res) => {
5   res.writeHead(200, { 'Content-Type': 'text/plain' });
6   res.end('hello world\n');
7 });
8
9 // starts a simple http server locally on port 3000
10 server.listen(3000, '127.0.0.1', () => {
11   console.log('listening on 127.0.0.1:3000');
12 });
13
14 // run with 'node server.js'

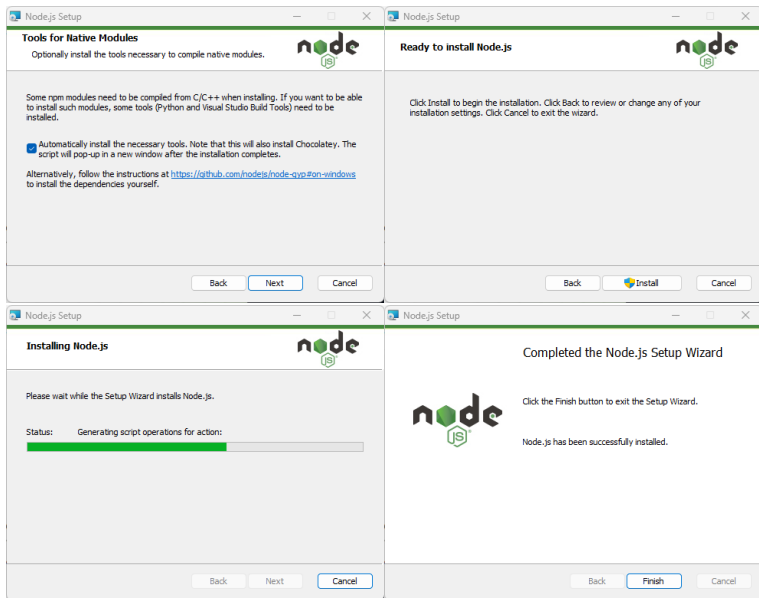
JavaScript Copy to clipboard
```

Learn more what Node.js is able to offer with our Learning materials.

[Trademark Policy](#) [Privacy Policy](#) [Code of Conduct](#) [Security Policy](#)[© OpenJS Foundation](#) [🔄](#) [📄](#) [🔗](#) [🌐](#) [📱](#)

Primer necessitem el programa npm per l'instal·lació d'Expo que serà el programa que carregarà els nostres projectes de React Native y gestionará la seva execució als dispositius.





Ara iniciem a la terminal l'instal·lació d'Expo amb el següent comandament.

```
C:\Users\ies>npm i -g expo-cli
```

```
C:\Users\iesrg> npm install --no-audit --no-fund --no-progress --no-verbose --no-dry-run --no-color --no-silent --no-shrinkwrap --no-package-lock --no-lockfile --no-save --no-force --no-global --no-root --no-bin-links --no-linker --no-peer --no-optional --no-strict-peer-deps --no-strict-plug
```

```
C:\Users\iesrg> npm install -g expo-cli
```

```
npm WARN deprecated stable@0.1.8: Modern JS already guarantees Array#sort() is a stable sort, so this library is deprecated. See the compatibility table on MDN: https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Global_Objects/Array/sort#browser_compatibility
```

```
npm WARN deprecated @babel/plugin-proposal-private-property-in-object@7.16.7: This proposal has been merged to the ECMAScript standard and thus this plugin is no longer maintained. Please use @babel/plugin-transform-private-property-in-object instead.
```

```
npm WARN deprecated @babel/plugin-proposal-private-methods@7.16.7: This proposal has been merged to the ECMAScript standard and thus this plugin is no longer maintained. Please use @babel/plugin-transform-private-methods instead.
```

```
npm WARN deprecated @babel/plugin-proposal-nullish-coalescing-operator@7.18.6: This proposal has been merged to the ECMAScript standard and thus this plugin is no longer maintained. Please use @babel/plugin-transform-nullish-coalescing-operator instead.
```

```
npm WARN deprecated @babel/plugin-proposal-numeric-separator@7.18.6: This proposal has been merged to the ECMAScript standard and thus this plugin is no longer maintained. Please use @babel/plugin-transform-numeric-separator instead.
```

```
npm WARN deprecated @babel/plugin-proposal-optional-catch-binding@7.18.6: This proposal has been merged to the ECMAScript standard and thus this plugin is no longer maintained. Please use @babel/plugin-transform-optional-catch-binding instead.
```

```
npm WARN deprecated @babel/plugin-proposal-class-properties@7.18.6: This proposal has been merged to the ECMAScript standard and thus this plugin is no longer maintained. Please use @babel/plugin-transform-class-properties instead.
```

```
npm WARN deprecated @babel/plugin-proposal-optional-chaining@7.21.0: This proposal has been merged to the ECMAScript standard and thus this plugin is no longer maintained. Please use @babel/plugin-transform-optional-chaining instead.
```

```
npm WARN deprecated @babel/plugin-proposal-async-generator-functions@7.20.7: This proposal has been merged to the ECMAScript standard and thus this plugin is no longer maintained. Please use @babel/plugin-transform-async-generator-functions instead.
```

```
npm WARN deprecated @babel/plugin-proposal-export-named-from@7.18.9: This proposal has been merged to the ECMAScript standard and thus this plugin is no longer maintained. Please use @babel/plugin-transform-export-named-from instead.
```

```
npm WARN deprecated @babel/plugin-proposal-object-rest-spread@7.20.7: This proposal has been merged to the ECMAScript standard and thus this plugin is no longer maintained. Please use @babel/plugin-transform-object-rest-spread instead.
```

```
npm WARN deprecated uuid@3.4.0: Please upgrade to version 7 or higher. Older versions may use Math.random() in certain circumstances, which is known to be problematic. See https://v8.dev/blog/math-random for details.
```

```
npm WARN deprecated uuid@3.4.0: Please upgrade to version 7 or higher. Older versions may use Math.random() in certain circumstances, which is known to be problematic. See https://v8.dev/blog/math-random for details.
```

```
npm WARN deprecated rimraf@2.7.1: Rimraf versions prior to v4 are no longer supported
```

```
npm WARN deprecated uuid@3.4.0: Please upgrade to version 7 or higher. Older versions may use Math.random() in certain circumstances, which is known to be problematic. See https://v8.dev/blog/math-random for details.
```

```
npm WARN deprecated osenv@0.1.5: This package is no longer supported.
```

```
npm WARN deprecated inflight@1.0.6: This module is not supported, and leaks memory. Do not use it. Check out lru-cache if you want a good and tested way to coalesce async requests by a key value, which is much more comprehensive and powerful.
```

```
npm WARN deprecated @babel/plugin-proposal-private-property-in-object@7.16.7: This proposal has been merged to the ECMAScript standard and thus this plugin is no longer maintained. Please use @babel/plugin-transform-private-property-in-object instead.
```

```
npm WARN deprecated rimraf@2.8.0: Rimraf versions prior to v4 are no longer supported
```

```
npm WARN deprecated rimraf@2.8.0: Rimraf versions prior to v4 are no longer supported
```

```
npm WARN deprecated glob@6.0.4: Glob versions prior to v9 are no longer supported
```

```
npm WARN deprecated glob@7.1.6: Glob versions prior to v9 are no longer supported
```

```
added 1299 packages in 28s
```

```
171 packages are looking for funding  
run `npm fund` for details
```

```
C:\Users\iesr>
```

Podem actualitzar alguns apartats d'npm per solventar aquests warning. Pot ser diferent als vostres ordinadors.

```
C:\Users\ies>npm install -g uuid@latest  
  
changed 1 package in 430ms  
  
1 package is looking for funding  
  run 'npm fund' for details  
  
C:\Users\ies>npm install -g rimraf@latest  
  
changed 41 packages in 734ms  
  
14 packages are looking for funding  
  run 'npm fund' for details  
  
C:\Users\ies>npm ls -g --depth=0  
C:\Users\ies\AppData\Roaming\npm  
+-- expo-cli@6.3.12  
+-- npm@10.9.0  
+-- rimraf@6.0.1  
`-- uuid@10.0.0
```

```
C:\Users\ies>npm audit  
found 0 vulnerabilities
```

Això ens donarà un informe de les vulnerabilitats actuals. El resultat típic inclourà:

- Paquets afectats.
- Severitat de la vulnerabilitat.
- Versió dels paquets que tenen els problemes.
- Descripció dels problemes y l'enllaç a més informació.
- Solució suggerida, como actualitzar el paquet.

La comanda per crear projecte en React Native:

```
C:\Users\ies>npx create-expo-app firstapp
Need to install the following packages:
create-expo-app@3.0.0
Ok to proceed? (y) y
```

✓ Your project is ready!

To run your project, navigate to the directory and run one of the following npm commands.

```
- cd firstapp
- npm run android
- npm run ios # you need to use macOS to build the iOS project - use the Expo app if you need to do iOS development without a Mac
- npm run web
```

```
C:\Users\ies>
```



Situeu-vos sobre la carpeta on s'ha creat el projecte React Native i ara ho iniciem expo.

Executant l'inici d'Expo amb aquesta comanda:

```
C:\Users\ies\firstapp>npx expo start
Starting project at C:\Users\ies\firstapp
Starting Metro Bundler
```



```
> Metro waiting on exp://192.168.1.19:8081
> Scan the QR code above with Expo Go (Android) or the Camera app (iOS)

> Web is waiting on http://localhost:8081

> Using Expo Go
> Press s | switch to development build

> Press a | open Android
> Press w | open web

> Press j | open debugger
> Press r | reload app
> Press m | toggle menu
> Press o | open project code in your editor

> Press ? | show all commands

Logs for your project will appear below. Press Ctrl+C to exit.
```

Problemes amb l'Editor d'Expo

Potser que Expo us intenti carregar el projecte amb el bloc de notes o el notepad en comptes del VisualStudioCode, el programa adequat. Intenteu el següent.

```
C:\Windows\system32\cmd.exe
C:\Users\ies>code
C:\Users\ies>setx EDITOR "code"
CORRECTO: se guardó el valor especificado.

C:\Users\ies>expo config editor code
WARNING: The legacy expo-cli does not support Node v17. Migrate to the new local Expo CLI: https://blog.expo.dev/the-new-expo-cli-f4259d8e3421.

The global expo-cli package has been deprecated.
The new Expo CLI is now bundled in your project in the expo package.
Learn more: https://blog.expo.dev/the-new-expo-cli-f4259d8e3421.
To use the local CLI instead (recommended in SDK 46 and higher), run:
  npx expo <command>

The expected package.json path: C:\Users\ies\editor\package.json does not exist
C:\Users\ies>
```

```
{ } state.json x
C: > Users > ies > .expo > { } state.json > ...
1  {
2    "analyticsDeviceId": "fc933
3    "PATH": "C:\\Python312\\Scr
4    "uuid": "ae6d0242-7908-4ec2
5    "editor": "code"
6  }
7
```


Amb el QR de la terminal d'Expo podem executar l'aplicació al nostre dispositiu mòbil amb l'aplicació d'Expo que trobareu a la PlayStore:

```
C:\Users\ies\firstapp>npx expo start
Starting project at C:\Users\ies\firstapp
Starting Metro Bundler
```



```
> Metro waiting on exp://192.168.1.19:8081
> Scan the QR code above with Expo Go (Android) or the Camera app (iOS)
> Web is waiting on http://localhost:8081

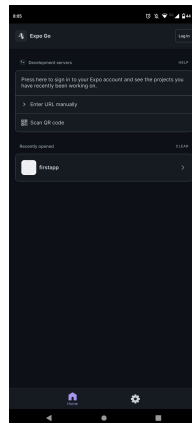
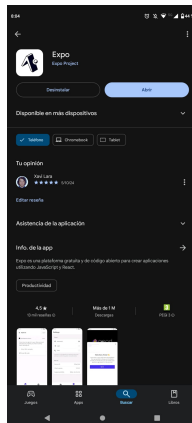
> Using Expo Go
> Press s | switch to development build

> Press a | open Android
> Press w | open web

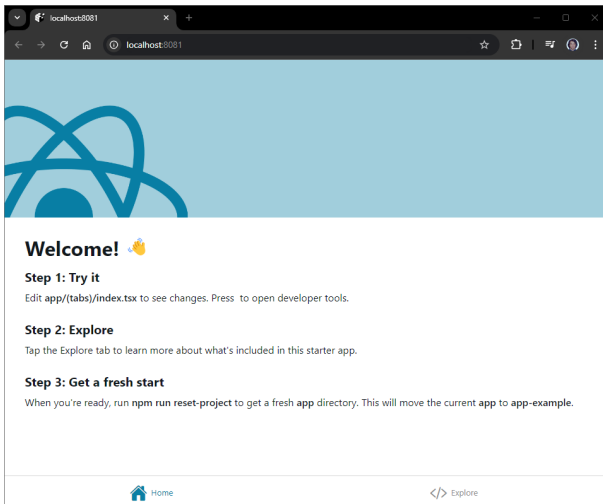
> Press j | open debugger
> Press r | reload app
> Press m | toggle menu
> Press o | open project code in your editor

> Press ? | show all commands

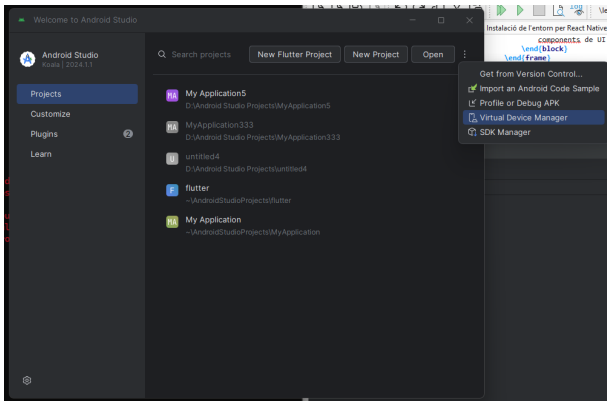
Logs for your project will appear below. Press Ctrl+C to exit.
```



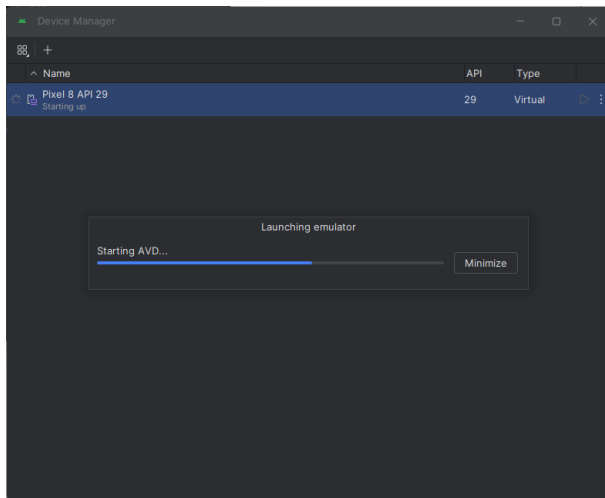
Entrant **w** a la terminal podem executar l'aplicació al explorador web del mateix ordinador.



Inicieu Android Studio sense obrir cap projecte Android i accediu a l'opció de **Virtual Device Manager**.



S'obrirà el mateix menu dels AVD utilitzats als projectes d'Android Studio. Engeguem algú:



Un cop engegat amb la comanda **a** o **swift+a** podreu seleccionar aquest emulador per executar l'aplicació del projecte de React Native.

